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Assignment 5.Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and ridge, Lasso regression models. 5. Evaluate the models and compare their respective scores like R2, RMSE, etc. Dataset link: https://www.kaggle.com/datasets/yasserh/uber-fares-dataset

```
from google.colab import files

uploaded = files.upload()
```

Choose files uber.csv
uber.csv(text/csv) - 23458139 bytes, last modified: 04/02/2026 - 100% done
Saving uber.csv to uber.csv

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error, mean_absolute_error,r2_score
```

```
df = pd.read_csv('uber.csv')
df
```

	Unnamed: 0	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dro
0	24238194	2015-05-07 19:52:06.0000003	7.5	2015-05-07 19:52:06 UTC	-73.999817	40.738354	-73.999512	
1	27835199	2009-07-17 20:04:56.0000002	7.7	2009-07-17 20:04:56 UTC	-73.994355	40.728225	-73.994710	
2	44984355	2009-08-24 21:45:00.00000061	12.9	2009-08-24 21:45:00 UTC	-74.005043	40.740770	-73.962565	
3	25894730	2009-06-26 08:22:21.0000001	5.3	2009-06-26 08:22:21 UTC	-73.976124	40.790844	-73.965316	
4	17610152	2014-08-28 17:47:00.000000188	16.0	2014-08-28 17:47:00 UTC	-73.925023	40.744085	-73.973082	
...	...	...	...	...	...	...	...	
199995	42598914	2012-10-28 10:49:00.00000053	3.0	2012-10-28 10:49:00 UTC	-73.987042	40.739367	-73.986525	
199996	16382965	2014-03-14 01:09:00.0000008	7.5	2014-03-14 01:09:00 UTC	-73.984722	40.736837	-74.006672	
199997	27804658	2009-06-29 00:42:00.00000078	30.9	2009-06-29 00:42:00 UTC	-73.986017	40.756487	-73.858957	
199998	20259894	2015-05-20 14:56:25.0000004	14.5	2015-05-20 14:56:25 UTC	-73.997124	40.725452	-73.983215	
199999	11951496	2010-05-15 04:08:00.00000076	14.1	2010-05-15 04:08:00 UTC	-73.984395	40.720077	-73.985508	

200000 rows × 9 columns

```
df.shape
```

```
(200000, 9)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 9 columns):
```

```
 #   Column             Non-Null Count   Dtype
---  ------             --------------   -----
 0   Unnamed: 0         200000 non-null  int64
 1   key                200000 non-null  object
 2   fare_amount        200000 non-null  float64
 3   pickup_datetime    200000 non-null  object
 4   pickup_longitude   200000 non-null  float64
 5   pickup_latitude    200000 non-null  float64
 6   dropoff_longitude  199999 non-null  float64
 7   dropoff_latitude   199999 non-null  float64
 8   passenger_count    200000 non-null  int64
dtypes: float64(5), int64(2), object(2)
memory usage: 13.7+ MB
```

```
df.drop(['Unnamed: 0', 'key'], axis=1, inplace=True)
```

```
df.isnull().sum()
df.dropna(inplace=True)
```

```
df = df[(df['fare_amount'] > 0) & (df['passenger_count'] > 0)].copy()
```

Uber fares can vary based on time-related factors such as:

time of day

month

year

Since machine learning models cannot directly understand datetime values, we convert the datetime column into useful numerical features.

```
df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])

df['hour'] = df['pickup_datetime'].dt.hour
df['day'] = df['pickup_datetime'].dt.day
df['month'] = df['pickup_datetime'].dt.month
df['year'] = df['pickup_datetime'].dt.year

df.drop('pickup_datetime', axis=1, inplace=True)
```

Feature Engineering – Distance Calculation

The distance between the pickup and drop-off locations is calculated using the Haversine formula.

Haversine Formula: The Haversine formula is a mathematical method used to compute the shortest distance between two points on the Earth's surface using their latitude and longitude coordinates. This distance is known as the great-circle distance.

```
def haversine(lat1, lon1, lat2, lon2):
    R = 6371  # Radius of Earth in km
    lat1, lon1, lat2, lon2 = map(np.radians, [lat1, lon1, lat2, lon2])

    dlat = lat2 - lat1
    dlon = lon2 - lon1

    a = np.sin(dlat/2)**2 + np.cos(lat1)*np.cos(lat2)*np.sin(dlon/2)**2
    c = 2 * np.arcsin(np.sqrt(a))

    return R * c

df['distance_km'] = haversine(
    df['pickup_latitude'], df['pickup_longitude'],
    df['dropoff_latitude'], df['dropoff_longitude']
)
```

```
df['distance_km']
```

	distance_km
0	1.683323
1	2.457590
2	5.036377
3	1.661683
4	4.475450
...	...
199995	0.112210
199996	1.875050
199997	12.850319
199998	3.539715
199999	5.417783

199269 rows × 1 columns

dtype: float64

```
df
```

	fare_amount	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	passenger_count	hour	day	mon
0	7.5	-73.999817	40.738354	-73.999512	40.723217	1	19	7	
1	7.7	-73.994355	40.728225	-73.994710	40.750325	1	20	17	
2	12.9	-74.005043	40.740770	-73.962565	40.772647	1	21	24	
3	5.3	-73.976124	40.790844	-73.965316	40.803349	3	8	26	
4	16.0	-73.925023	40.744085	-73.973082	40.761247	5	17	28	
...	...	...	...	...	...	...	...	...	
199995	3.0	-73.987042	40.739367	-73.986525	40.740297	1	10	28	
199996	7.5	-73.984722	40.736837	-74.006672	40.739620	1	1	14	
199997	30.9	-73.986017	40.756487	-73.858957	40.692588	2	0	29	
199998	14.5	-73.997124	40.725452	-73.983215	40.695415	1	14	20	
199999	14.1	-73.984395	40.720077	-73.985508	40.768793	1	4	15	

199269 rows × 11 columns

II. Identify outliers: Outlier detection is used to identify and handle abnormal data points that differ significantly from the rest of the dataset.

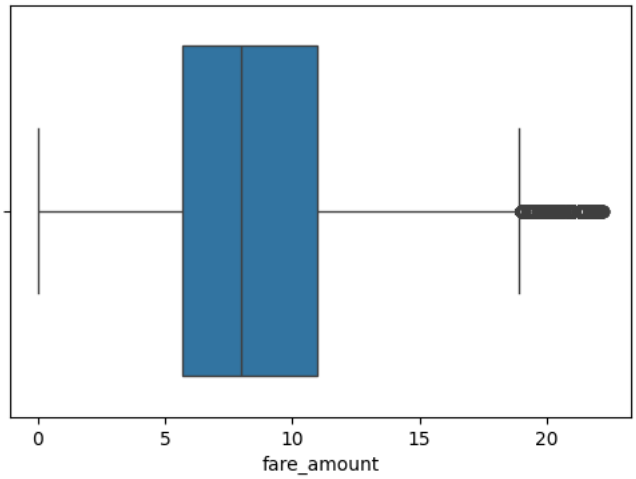
In Uber fare prediction, extreme fare values can negatively affect model performance.

IQR:Interquartile Range (IQR) help detect and remove these outliers, improving model accuracy and reliability.

```
Q1 = df['fare_amount'].quantile(0.25)
Q3 = df['fare_amount'].quantile(0.75)
IQR = Q3 - Q1

df = df[(df['fare_amount'] >= Q1 - 1.5*IQR) &
        (df['fare_amount'] <= Q3 + 1.5*IQR)]
```

A box plot is used to visualize the distribution of the fare amount and identify outliers. Values outside the whiskers indicate unusually high or low fares.

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['fare_amount'])
plt.show()
```



III. Check Coorelation Correlation is a statistical measure that shows the strength and direction of the relationship between two variables.

In our dataset, correlation shows how factors like distance and time influence Uber fares.

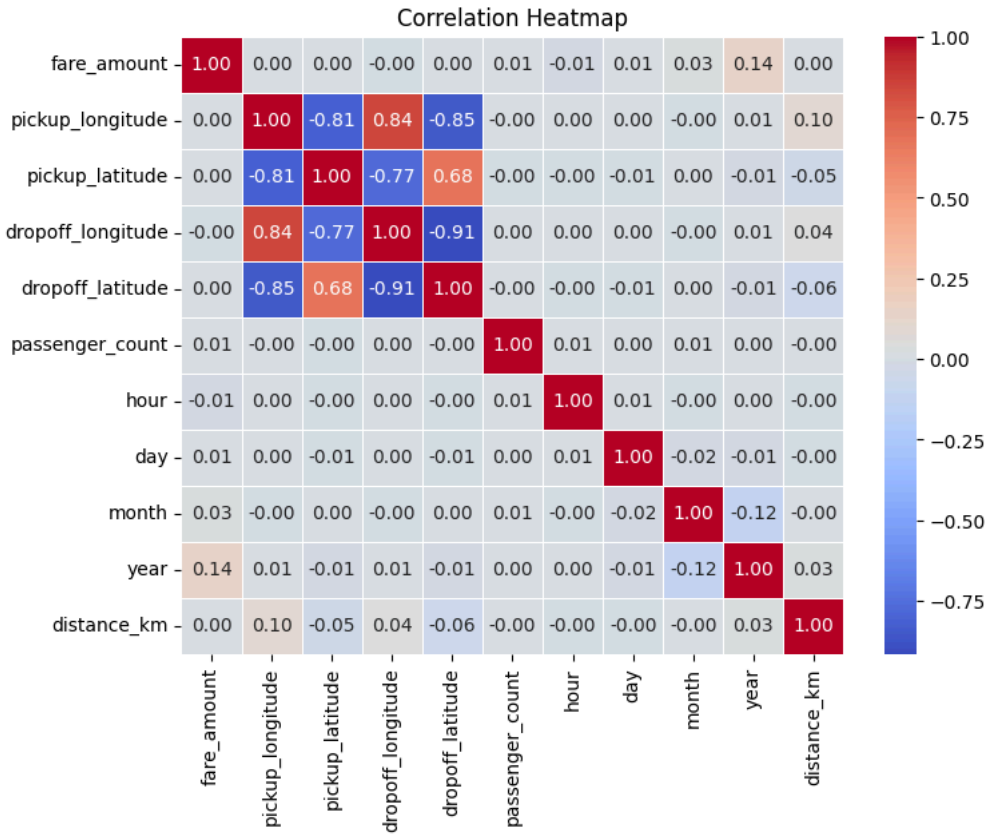
It assists in selecting important features and improving model performance.

```
corr=df.corr()
corr
```

	fare_amount	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	passenger_count	
fare_amount	1.000000	0.000125	0.002826	-0.002534	0.000715	0.011995	-0.01
pickup_longitude	0.000125	1.000000	-0.811435	0.835878	-0.850520	-0.000582	0.00
pickup_latitude	0.002826	-0.811435	1.000000	-0.766797	0.683972	-0.001742	-0.00
dropoff_longitude	-0.002534	0.835878	-0.766797	1.000000	-0.913666	0.000271	0.00
dropoff_latitude	0.000715	-0.850520	0.683972	-0.913666	1.000000	-0.000911	-0.00
passenger_count	0.011995	-0.000582	-0.001742	0.000271	-0.000911	1.000000	0.01
hour	-0.013304	0.003668	-0.004779	0.004220	-0.003373	0.013306	1.00
day	0.007128	0.003436	-0.006933	0.003941	-0.006237	0.003378	0.00
month	0.026188	-0.003842	0.004069	-0.003292	0.003317	0.009190	-0.00
year	0.135630	0.009803	-0.009843	0.007351	-0.009972	0.001532	0.00
distance_km	0.003022	0.096633	-0.047525	0.038328	-0.064035	-0.000153	-0.00

Next steps: Generate code with corr New interactive sheet

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)

plt.title("Correlation Heatmap")
plt.show()
```




```
data={
    "Study_Hours": [2,5,3,6,4,7,1,8,2,5],
    "Attendance_Percentage": [75,90,65,95,80,98,60,85,55,88],
    "Sleep_Hours": [6,7,5,8,6,7,4,8,5,7],
    "Final_Score": [58,88,60,92,75,95,50,90,55,85]
    }

df1=pd.DataFrame(data)
df1
```

	Study_Hours	Attendance_Percentage	Sleep_Hours	Final_Score
0	2	75	6	58
1	5	90	7	88
2	3	65	5	60
3	6	95	8	92
4	4	80	6	75
5	7	98	7	95
6	1	60	4	50
7	8	85	8	90
8	2	55	5	55
9	5	88	7	85

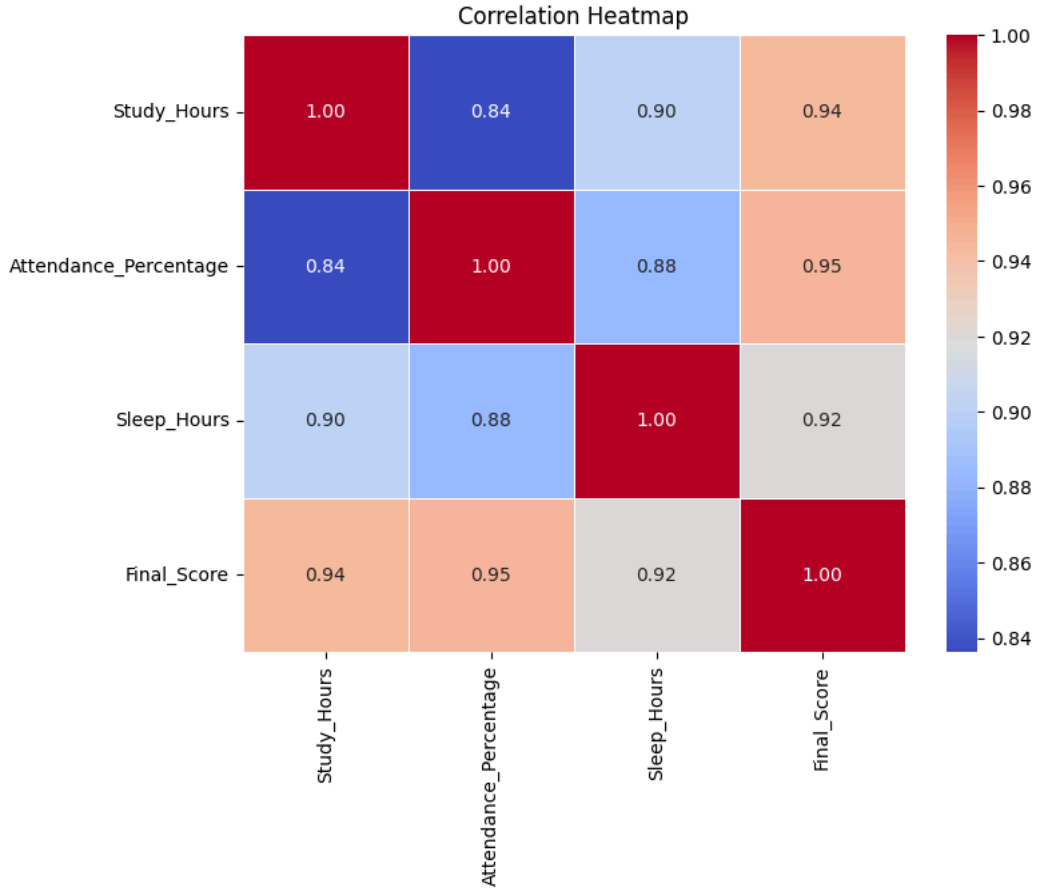
Next steps: Generate code with df1 New interactive sheet

```
corr=df1.corr()
corr
```

	Study_Hours	Attendance_Percentage	Sleep_Hours	Final_Score
Study_Hours	1.000000	0.836283	0.901962	0.944133
Attendance_Percentage	0.836283	1.000000	0.883319	0.945322
Sleep_Hours	0.901962	0.883319	1.000000	0.920092
Final_Score	0.944133	0.945322	0.920092	1.000000

Next steps: Generate code with corr New interactive sheet

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    fmt='.2f',
    linewidths=0.5
)

plt.title("Correlation Heatmap")
plt.show()
```



Correlation between two specific columns such as study hours and final score

```
corr_col=df1['Study_Hours'].corr(df1['Final_Score'])
corr_col
```

```
np.float64(0.9441329957211195)
```

```
import numpy as np
x=[1,2,3,4,5,6]
y=[10,23,67,32,13,56]
corr = np.corrcoef(x,y)
print(corr)
```

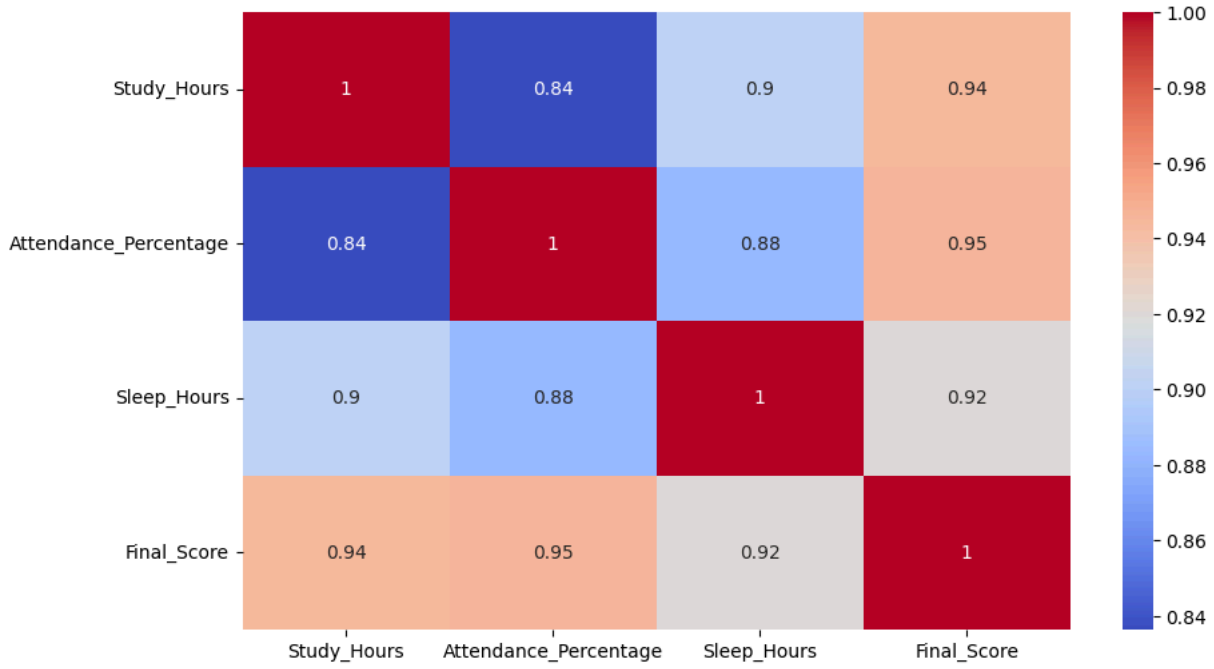
```
[[1.         0.37859076]
 [0.37859076 1.        ]]
```

```
corr = df1.dropna().corr()
print(corr)
```

```
                      Study_Hours  Attendance_Percentage  Sleep_Hours  \
Study_Hours              1.000000               0.836283     0.901962
Attendance_Percentage    0.836283               1.000000     0.883319
Sleep_Hours              0.901962               0.883319     1.000000
Final_Score              0.944133               0.945322     0.920092

                      Final_Score
Study_Hours              0.944133
Attendance_Percentage    0.945322
Sleep_Hours              0.920092
```

```
Final_Score                1.000000
```

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))
sns.heatmap(df1.corr(), annot=True, cmap='coolwarm')
plt.show()
```




```
corr=df1.corr
corr_value=df1
```

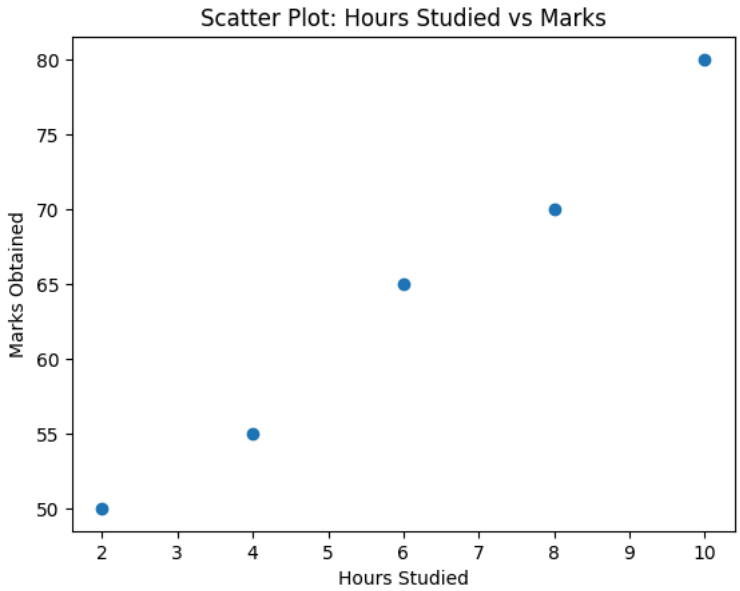
Strong correlation: Pickup and drop-off latitude/longitude are strongly correlated with each other.

Moderate correlation: Fare amount and distance show a noticeable positive correlation.

Weak correlation: Fare amount with passenger count, hour, day, month, and year shows weak correlation.

IV. Implement linear regression on the Hour and marks dataset also on the Uber dataset

```
import matplotlib.pyplot as plt

# Data
hours = [2, 4, 6, 8, 10]
marks = [50, 55, 65, 70, 80]

# Scatter plot
plt.scatter(hours, marks)
plt.xlabel("Hours Studied")
plt.ylabel("Marks Obtained")
plt.title("Scatter Plot: Hours Studied vs Marks")

plt.show()
```




```
import pandas as pd
data={
     'Hours_Studied': [2, 4, 6, 8, 10],
     'Marks': [50, 55, 65, 75, 85]
}
df2=pd.DataFrame(data)
df2
```

	Hours_Studied	Marks
0	2	50
1	4	55
2	6	65
3	8	75
4	10	85

Next steps: Generate code with df2 New interactive sheet

```
from sklearn.linear_model import LinearRegression

X = df2[['Hours_Studied']]
y = df2['Marks']

model = LinearRegression()
model.fit(X, y)

print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
```

```
Intercept: 39.0
Slope: 4.5
```

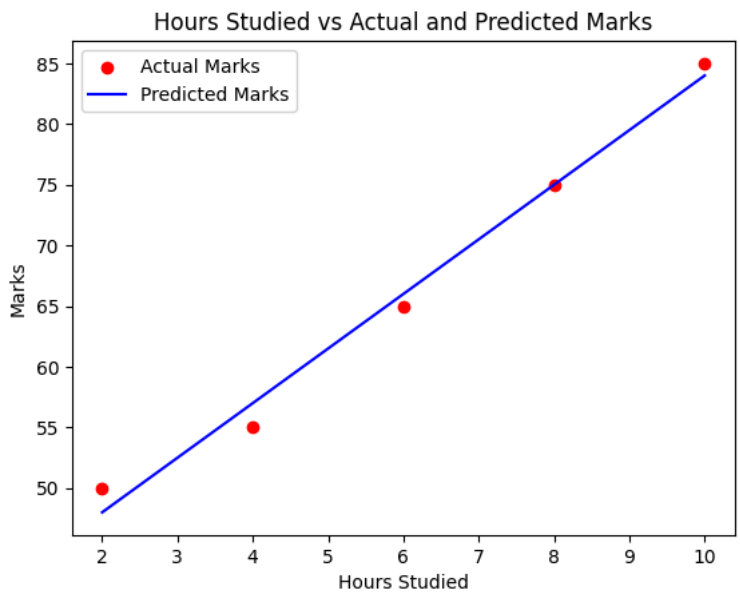
```
df2['predicted_values']=model.predict(X)
print(df2)
```

```
   Hours_Studied  Marks  predicted_values
0              2     50              48.0
1              4     55              57.0
2              6     65              66.0
3              8     75              75.0
4             10     85              84.0
```

```
from sklearn.metrics import mean_squared_error, r2_score

mse = mean_squared_error(y, df2['predicted_values'])
r2 = r2_score(y, df2['predicted_values'])

print("MSE:", mse)
print("R² Score:", r2)
```

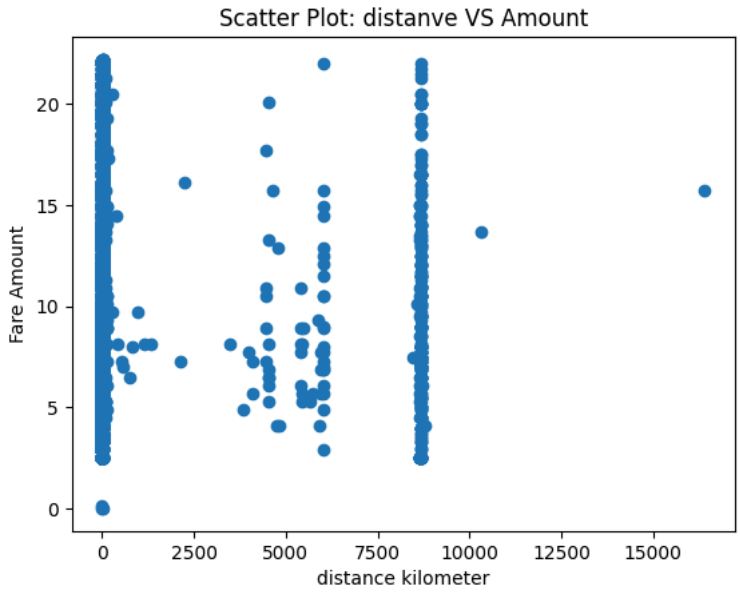
```
MSE: 2.0
R² Score: 0.9878048780487805
```

```
plt.figure()

# Actual values (Red scatter)
plt.scatter(df2['Hours_Studied'], df2['Marks'],
            color='red', label='Actual Marks')

# Predicted values (Blue line)
plt.plot(df2['Hours_Studied'], df2['predicted_values'],
         color='blue', label='Predicted Marks')

# Labels
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Hours Studied vs Actual and Predicted Marks")

# Legend
plt.legend()

plt.show()
```



Linear Regression on the Uber Dataset

```
import matplotlib.pyplot as plt

# Scatter plot
plt.scatter(df['distance_km'],df['fare_amount'])
plt.xlabel("distance kilometer")
plt.ylabel("Fare Amount")
plt.title("Scatter Plot: distanve VS Amount")

plt.show()
```




```
from sklearn.linear_model import LinearRegression

X = df[['distance_km']]
y = df['fare_amount']

model = LinearRegression()
model.fit(X, y)

print("Intercept:", model.intercept_)
print("Slope:", model.coef_[0])
```

```
Intercept: 8.942437479828127
Slope: 3.487982785271508e-05
```

```
df['predicted_values']=model.predict(X)
print(df)
```

```
        fare_amount  pickup_longitude  pickup_latitude  dropoff_longitude  \
0               7.5        -73.999817        40.738354         -73.999512
1               7.7        -73.994355        40.728225         -73.994710
2              12.9        -74.005043        40.740770         -73.962565
3               5.3        -73.976124        40.790844         -73.965316
4              16.0        -73.925023        40.744085         -73.973082
...             ...               ...              ...               ...
199994         12.0        -73.983070        40.760770         -73.972972
199995          3.0        -73.987042        40.739367         -73.986525
199996          7.5        -73.984722        40.736837         -74.006672
199998         14.5        -73.997124        40.725452         -73.983215
199999         14.1        -73.984395        40.720077         -73.985508

        dropoff_latitude  passenger_count  hour  day  month  year  \
0              40.723217                1    19    7      5  2015
1              40.750325                1    20   17      7  2009
2              40.772647                1    21   24      8  2009
3              40.803349                3     8   26      6  2009
4              40.761247                5    17   28      8  2014
...                  ...              ...   ...  ...    ...   ...
199994         40.754177                1    14   31      1  2014
199995         40.740297                1    10   28     10  2012
199996         40.739620                1     1   14      3  2014
199998         40.695415                1    14   20      5  2015
199999         40.768793                1     4   15      5  2010

        distance_km  predicted_values
0          1.683323          8.942496
1          2.457590          8.942523
2          5.036377          8.942613
3          1.661683          8.942495
4          4.475450          8.942594
...             ...               ...
199994     1.122878          8.942477
199995     0.112210          8.942441
199996     1.875050          8.942503
199998     3.539715          8.942561
199999     5.417783          8.942626

[182148 rows x 12 columns]
/tmp/ipython-input-1848995545.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  df['predicted_values']=model.predict(X)
```

```
mse = mean_squared_error(y, df['predicted_values'])
r2 = r2_score(y, df['predicted_values'])

print("MSE:", mse)
print("R² Score:", r2)
```